

LumenTrace

Simon Völkl , Erik Dombrowsky, and Simon Frey

Abstract—Background: Objective: Methods: Results: Conclusion:

Index Terms—Enter key words or phrases in alphabetical order, separated by commas.

THIS technical report documents the design, implementation, and evaluation of LumenTrace, a high-performance MCU-based LED racing simulator. [1], [2].

REFERENCES

- [1] S. Völkl, E. Dombrowsky, and S. Frey. (2026) Lumentrace github repository. [Online]. Available: <https://github.com/s-voelkl/LumenTrace>. GitHub Repository.
- [2] M. Kane and V. Edmondson, “Tire/road friction prediction: Introduction a simplified numerical tool based on contact modelling,” *Vehicle system dynamics*, vol. 60, no. 3, pp. 770–789, 2022.

The authors are with the Department of Electrical Engineering, Media, and Computer Science, Ostbayerische Technische Hochschule Amberg-Weiden, 92224 Amberg, Germany (e-mail: s.voelkl2@oth-aw.de; e.dombrowsky@oth-aw.de; s.frey@oth-aw.de).